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Exp No: 1

Azure DevOps Environment Setup

21/01/26

Aim:

To set up and access the azure devops environment by creating an organization through the azure portal.

Installation:

1. open your web browser and go to the azure website:

<https://azure.microsoft.com/en-us/get-started/azure-portal>

sign in using your microsoft account credentials.

If you don't have a microsoft account, you can create one here:

<https://signup.live.com/?lic=1>

Get started with Azure

Sign in

2. Azure home page

Azure services - Azure DevOps organizations

3. open devops environment in the Azure platform by typing Azure DevOps organizations in the search bar.

4. click on the my Azure DevOps organization link and create an organization and you should be taken to the Azure DevOps organization Home page.

Azure DevOps

my Azure DevOps organizations

Result: successfully accessed the Azure DevOps environment

and created a new organization through the azure portal.

Azure DevOps

Azure DevOps

Plan smarter, collaborate better, and ship faster with a set of modern dev services.

[View my organizations](#)

Microsoft Azure

[Search resources, services, and docs \(Ctrl-F\)](#)
[Copilot](#)

Azure services



Resources

Recent Favorite

Name

Type

No resources have been viewed recently

[View all resources](#)

28/01/26

Aims:

Produce:

~~Azusa~~ Douqns

add. in

daw.a.zura.cem / laithanfreshchina.kemburajagorod.2024.it

(a) after the organization is set up, you'll need to create your first project. This is where you'll begin to manage code, pipelines, work items and more.

(c) Enter the project name, description to provide more

Description: optionally, add a description to provide more

viability: choose whatever you want - the project to be private or public.



(d) once you've filled out the details, click **Create** to set up your first project.

Create new project

Project name: Weather App

visibility: Public

version control: Git

user team roles: [lalithsanthoshchinnakondragopal-2024-it-agile](#)
create

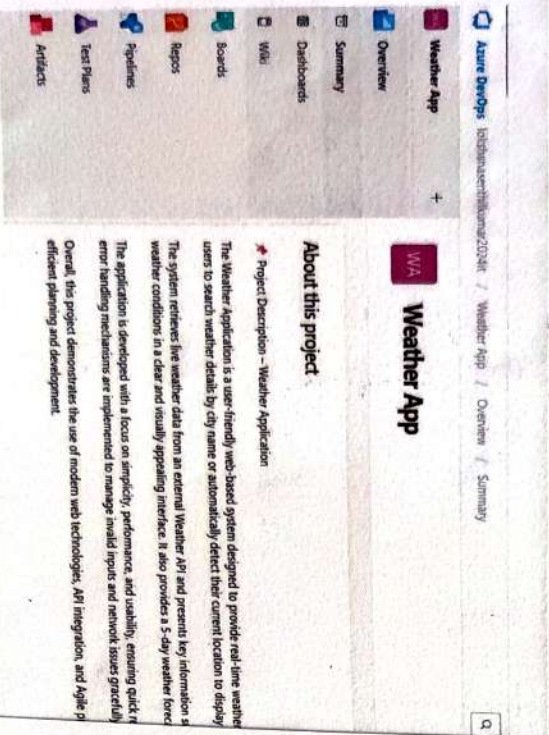
3. once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). click on the organization name, and you should be taken to the Azure DevOps organization home page.

LS Lalith Sathosh

lalithsanthoshchinnakondragopal-2024-it@triplabehmi.edu.in
dev.azure.com/lalithsanthoshchinnakondragopal-2024-it (member)

projects - weather App

4. Project dashboard.



About the project

The weather application is a user-friendly web app that provides real-time weather updates and forecasts for any city using an external API. It displays details like temperature, humidity, and weather conditions and includes features such as city-based search, location detection, error handling, and a simple interface.

Managing user stories

(a) Go to boards from the left menu to manage work items, backlogs, and sprints.

(b) On the ~~work~~ items page, click Add on + New work item, then select user story to create a new one.

boards → work items → + new work item → user story

Result:

Successfully created an Azure DevOps project with user story management and agile workflow setup.

Exp No: 3 Setting up Epics, Features & User Stories

11/02/26

for Project Planning

Aim:
to learn about how to create epics, user stories, features, backlog for your assigned projects.

Procedure:

create Epic, features, user stories, Task

As a user, I want to view accurate and real time weather information so that I can plan my daily activities accordingly.

1. Fill in Epics

Epic 1

weather information management

2. Fill in Features

2. city search feature

3. current weather display

4. weather display - Forecast Features

5. Location - based weather

6. error Handling & Alerts

7. user interface & experience

3. Fill in user story details

8. view weather by city name

9. view current weather details

10. view 5-day weather forecast

11. detect user location

Work Items		
My activity		
+ New Work Item Open in Queue Column Options Import Work Items Recycle Bin		
Filter by keyword		
ID	Title	Assigned to
1	Weather Information Management	Lokshana Senthikumar
2	City Search Feature	Unassigned
3	Current Weather Display	Unassigned
4	Weather Forecast Feature	Unassigned
5	Location-Based Weather	Unassigned
6	Error Handling & Alerts	Unassigned
7	User Interface & Experience	Unassigned
8	view weather by city name	jashwanth J
9	view current weather details	jashwanth J
10	view 5-day weather forecast	Lalith Santhosh
11	Detect user location	Lalith Santhosh
12	Handle invalid city input	joshi shree
13	Display user-friendly interface	joshi shree

12. Handle invalid city input.
13. Display user-friendly interface.
8. As a user, I want to search weather information by entering a city name so that I can know the weather conditions of any location.
9. As a user, I want to view the current temperature, humidity and weather conditions so that I can understand the present climate.
10. As a user, I want to see a 5-day weather forecast so that I can plan my future activities.
11. As a user, I want the application to automatically detect my location so that I receive weather updates without manual input.
12. As a user, I want to see an error message when an invalid city name is entered.
13. As a user, I want a simple and visually appealing interface so that the application is easy to use.

Result:

Thus, the creation of APIs, features, user stories, and tasks has been created successfully.

Exp No: 4

Sprint Planning

25/02/26

Aim:

To assign user story to specific sprint for the weather app project.

Sprint planning:

Sprint 1:

Taskboard

8. view weather by city

Assigned to: Jashwanth T

Iteration 1:

Taskboard

9. view current weather

Assigned to: Jashwanth T

Iteration 2:

Taskboard

10. view 5-day weather forecast

Assigned to: Lalith

11. detect user location

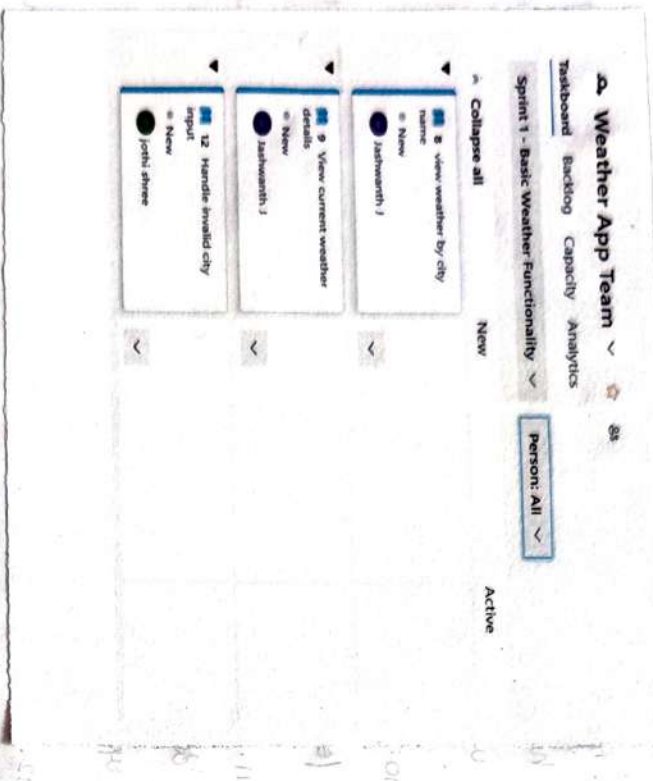
Assigned to: Lalith

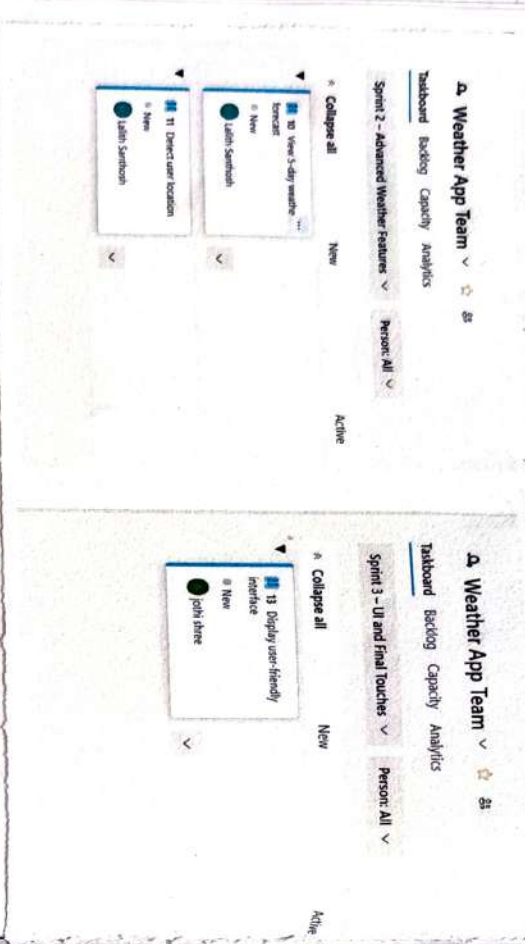
Iteration 3:

Task board

12. Handle invalid city input

Assigned to: Jothi shree





~~Result:~~
The sprints are created for the weather App project.

Exp No:5

11/03/26

Poker's Estimation

Aim:

To create poker estimation for the user stories - weather App project.

Poker Estimation:

Story Points measure:

- effort
- complexity
- time

This was Fibonacci scale: 1,2,3,5,8,13,...

Story points for weather App:

User Story	Story Points	Reason
Search weather by city	3	API call + input handling
View current weather details	2	Simple data display
View 5-day weather forecast	5	more data + UI
Detect user location	5	Permission + GPS
Handle invalid city input	2	Error Handling
Display user - friendly interface	3	UI styling

total = 20 story points

Records → work items → user story → story points → (1/2/3/5) → save



Result:
The estimation/story points is created for the project weather App using poker estimation.

Exp No: 6

Designing class and sequence diagrams for

Date: 18/3/20

Project ArchitectureAim:

To design a class diagram and sequence diagram for the given "Real estate portal" project.

6.1 class diagram:Definition:

A class diagram is a type of UML diagram that represents the structure of a system by showing its classes, attributes, methods, and relationships between them.

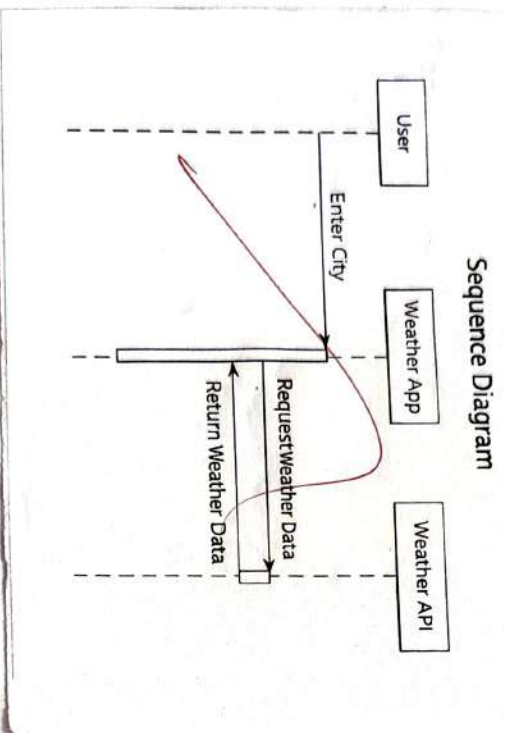
Features:

It helps to understand

- What objects exist in the system
- What data they store (attributes)
- What actions they perform (methods)
- How they are connected (associations, inheritance, etc)

6.2 sequence diagramDefinition:

A sequence diagram is a type of UML diagram that shows the interaction between different components of a system over time.

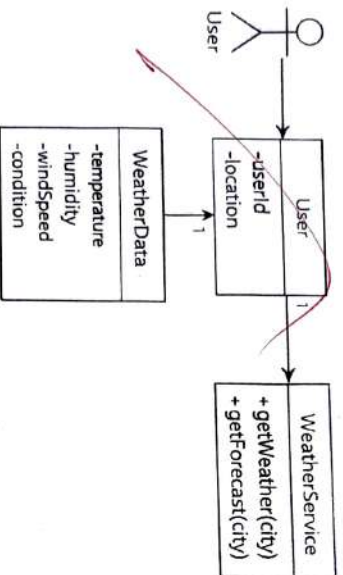


Features:

IT requirements:

- The sequence of messages exchanged
- The order in which operations occur
- The flow of control from one object to another.

Class Diagram



Result:

The class diagram and sequence diagram is designed successfully for the "weather app" project.

Exp no: 17 Designing Architectural and ER Diagrams

Date: 1/4/26

For project structure

Aim:

To design an Architectural diagram and ER diagram for the given project, "Weather App".

7A. Architectural Diagram

Definition:

Architectural Diagram shows the high-level structure of a system.

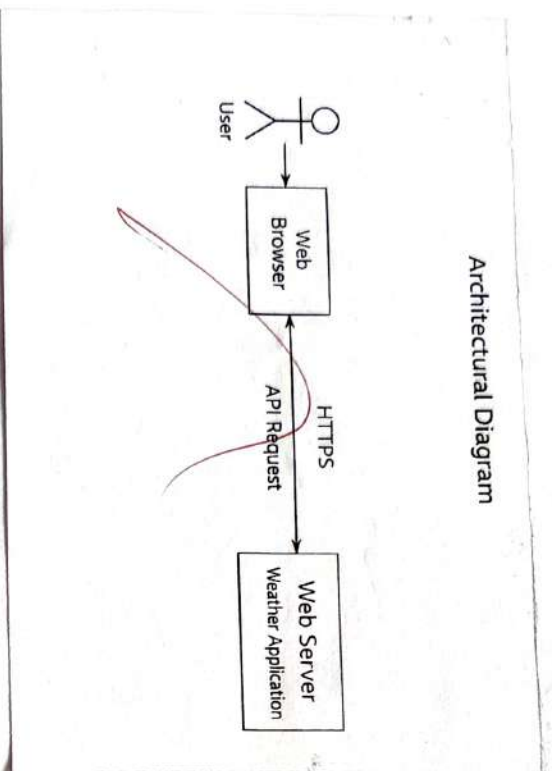
Features:

- Illustrates system components and their interactions
- Shows data/ process flows between components
- Helps to visualize overall architecture of the system.

7B ER Diagram

Definition:

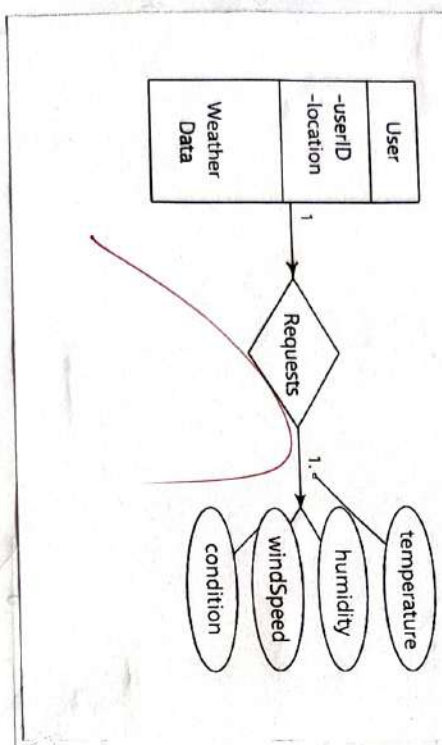
An ER diagram is a graphical representation of a database that shows entities, their attributes and the relationship between them.



Features:

- Identifies key entities
- Shows attributes of each entity
- Defines relationships between entities
(one-to-one, one-to-many, etc)

ER Diagram

Result:

The Architecture Diagram and ER diagram is designed successfully for the "Weather App" project.

8/04/26

8/04/26

Aim:

Aim: Test plans and test cases showcasing the happy path and errors scenarios in azure devops platform.

Test case design procedure:

1. Understand core features of the application
2. Design user interaction.
3. Design happy path test cases
4. Design error path test cases
5. Break down steps and expected results
6. Use clear naming and ID's
7. Separate Test Suite
8. Prioritize and Review

Test case 1: Happy path (Valid city search)

title: Verify weather details for valid city input

Steps:

Steps:		Expected result
S.No	Action	
1	open weather App	Application loads successfully
2	Enter valid city name	city name accepted
3	click search button	Request sent to API
4	view results	weather details displayed.

[illegible][illegible]

Pass

title:
verify error message for invalid city input

Steps:

Expected	Result:
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10
11	11
12	12
13	13
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99	99
100	100

Outcome:

Pass

~~Result:~~ ~~frup,~~ the test plans and test cases showcasing that happy path and error scenarios in Azure DevOps platform.

[illegible]

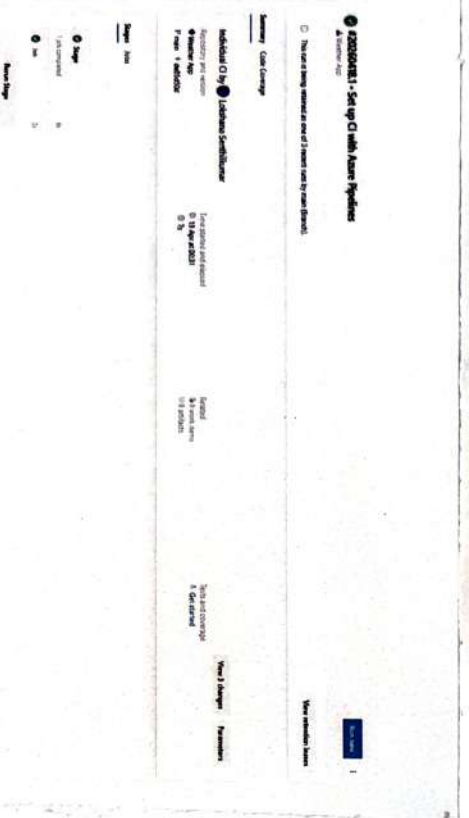
7. Create an azure head testing resource and

Azure Lead Testing:

Steps to create an Azure load testing resource:

- ## Steps to create and run a lead test

1. Click on HTTP requests > create
2. Basics tab - Test name, description
3. Local settings - test url
4. Click Review + create to start the test



Pipelines:

The pipelines will automatically install dependencies, run basic tests and publish artifacts. This ensures every commit triggers checks for stability and smooth deployment.

Steps:

1. Connect Github to azure devops
2. Create azure - pipelines.yml in your repo root
3. Include the pipeline tasks and install project dependencies
4. Run and monitor pipeline in Azure devops.

20260418.1 - setup CI with Azure Pipelines

Jobs in run # 20260418.1

Azure pipelines: ubuntu - latest

Duration: 28 - job initialized and completed.

yml file:

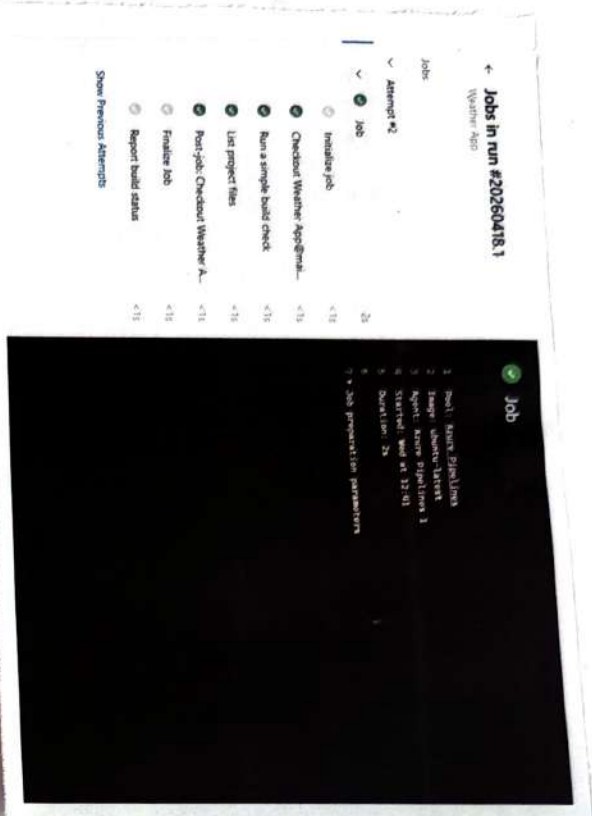
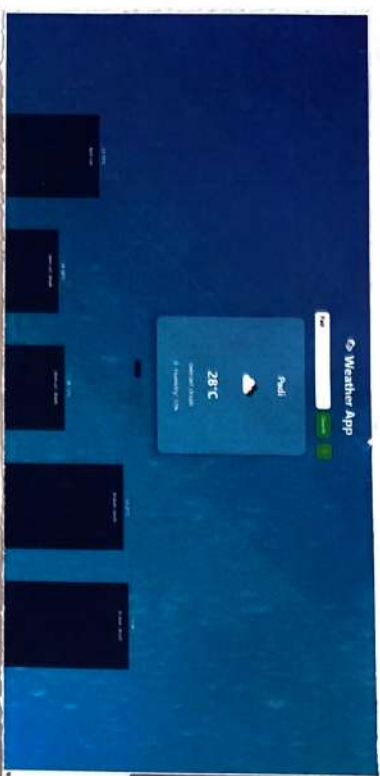
azure - pipelines.yml

triggers:

- main

pool:

vm image: 'ubuntu - latest'



Steps:

- script : echo - " creation App build successfully "
- displayName: "Run a simple build check"
- script : 1
- echo "checking files..."
- ls
- displayName: "List project files"

Result:

successfully created the azure load testing resource and executed a load test to assess the performance of the specified endpoint and also demonstrated pipelines in Azure devops.

Ex no:10 GitHub: Project Structure and Naming Conventions

25/04/2026

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate and extend the weather app project.

GitHub project structure:

rajith sandosh / weather-app

- * `gitignore` = remove API key & added `gitignore`
- * `index.html` = added HTML file of weather-app
- * `script.js` = Added the entire code for web app
- * `style.css` = Initial commit

Languages

Javascript - 67.8%

CSS - 18.4%

HTML - 13.8%

Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions making it easy for users and contributors to understand and navigate the codebase.

